Flux Hyperlocal Delivery Intelligence Platform

TEAM NO -4

MEMBERS:

ADARSH DUBEY-2023BCS0117

ROSHAN BINOJ-2023BCS0009

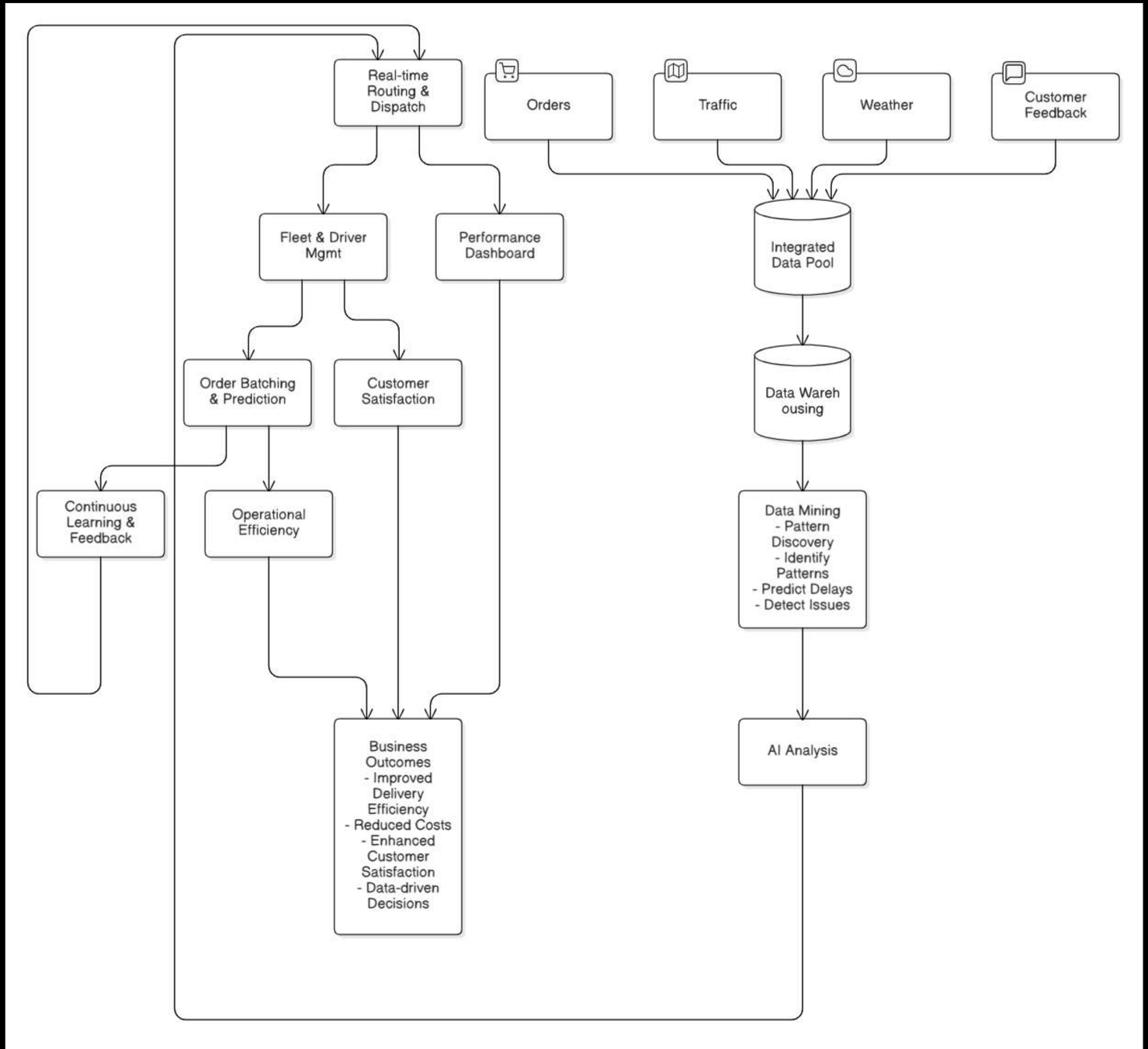
MOHAMMED SIRAJUDHEEN-2023BCS0132

HAFIZ FIROZ-2023BCS006

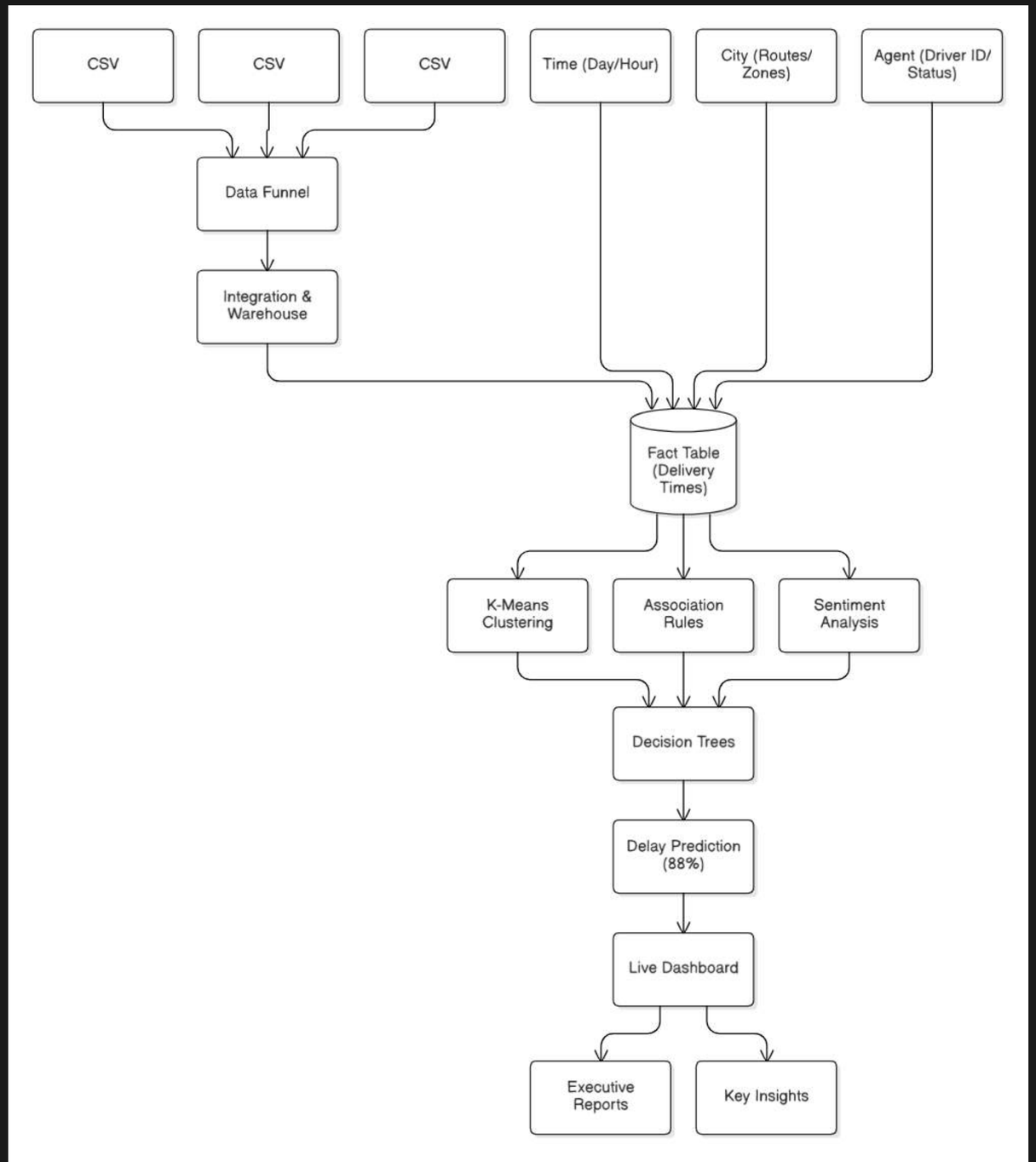
PROBLEM STATEMENT

- Hyperlocal delivery platforms face challenges due to fluctuating demand, traffic, weather conditions, and festival surges.
- These factors lead to delays, reduced customer satisfaction, and higher operational costs.
- Rider availability and inventory issues further impact delivery efficiency.
- Data is scattered across multiple sources and not structured for proper analysis.
- This makes it difficult for decision-makers to identify problems and analyze performance trends.

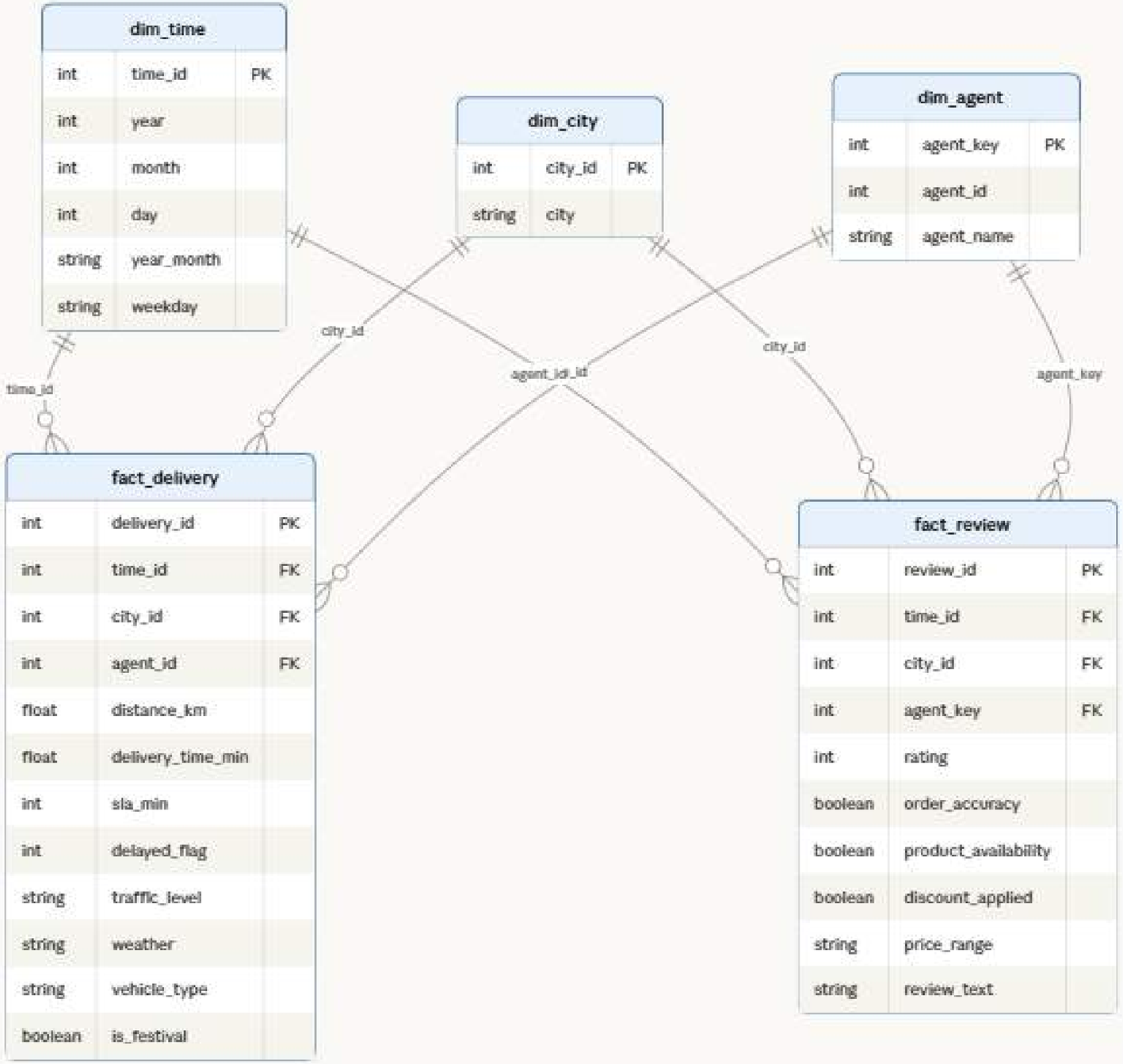
PROPOSED SOULTION



PROCESS FOLLOWED



Galaxy schema



Fact tables Dimension tables Primary key (PK) Foreign key (FK)

Technologies Used

TECHNOLOGIES

ETL: R, tidyverse, dplyr, tidyr

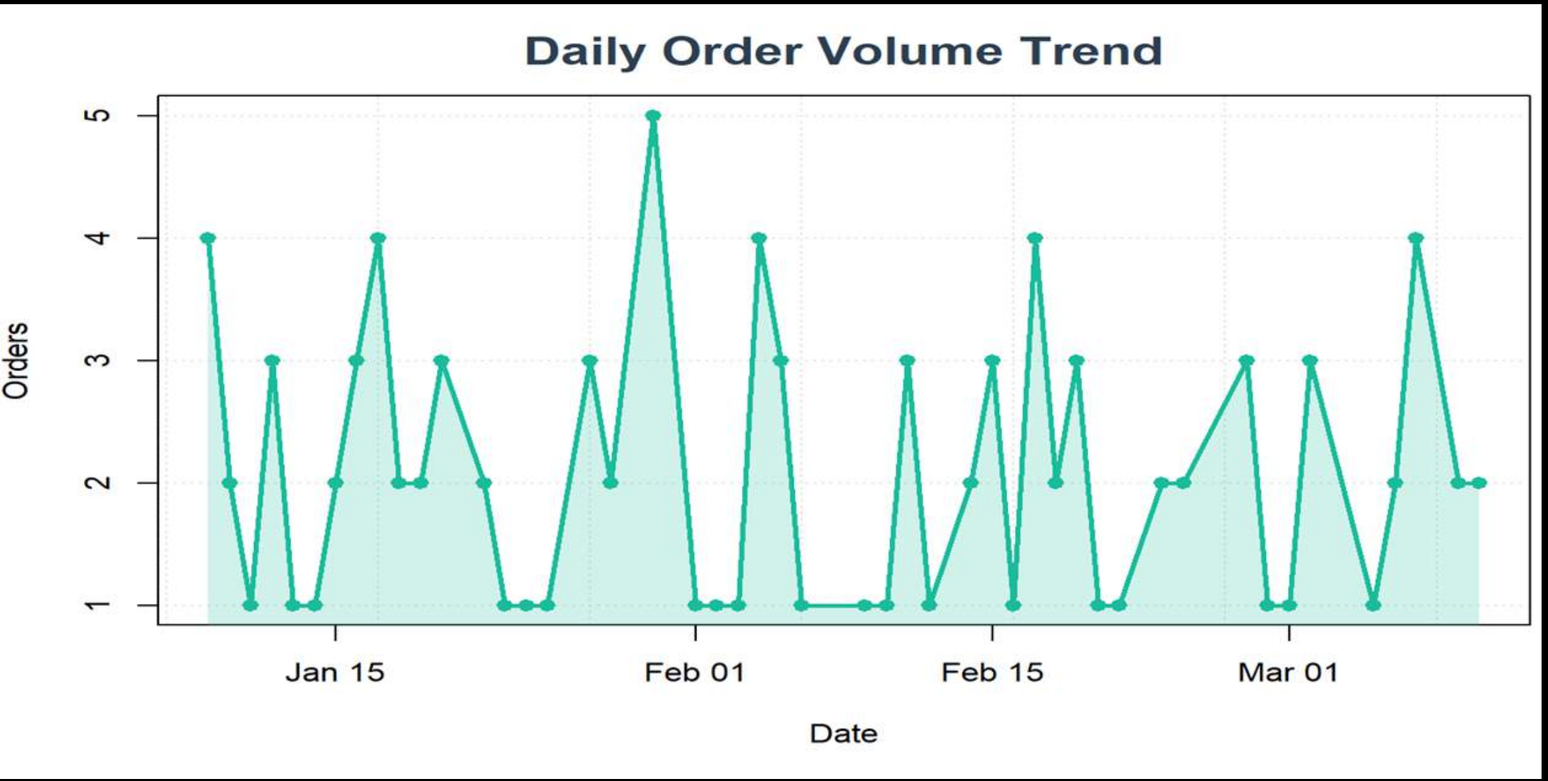
Warehouse: SQLite

Text Mining: tidytext, tm

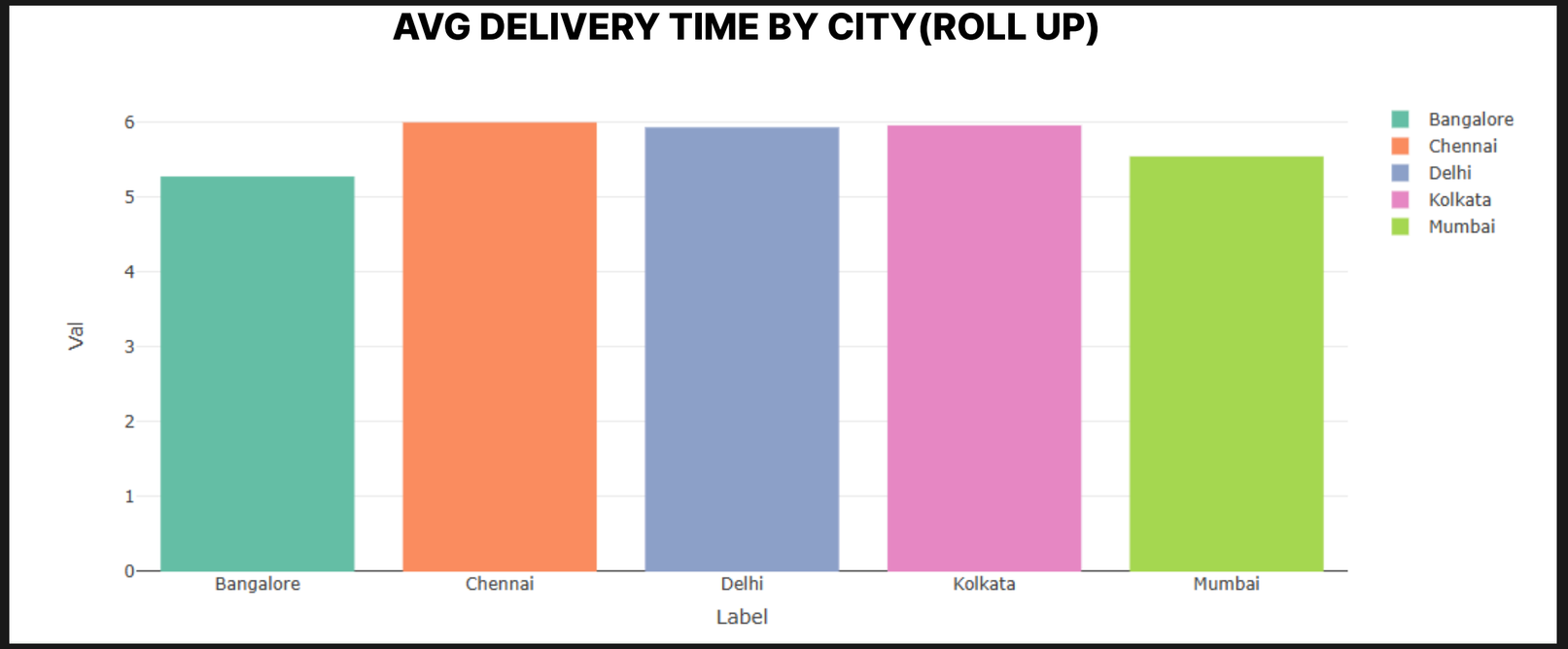
Visualization: ggplot2, plotly

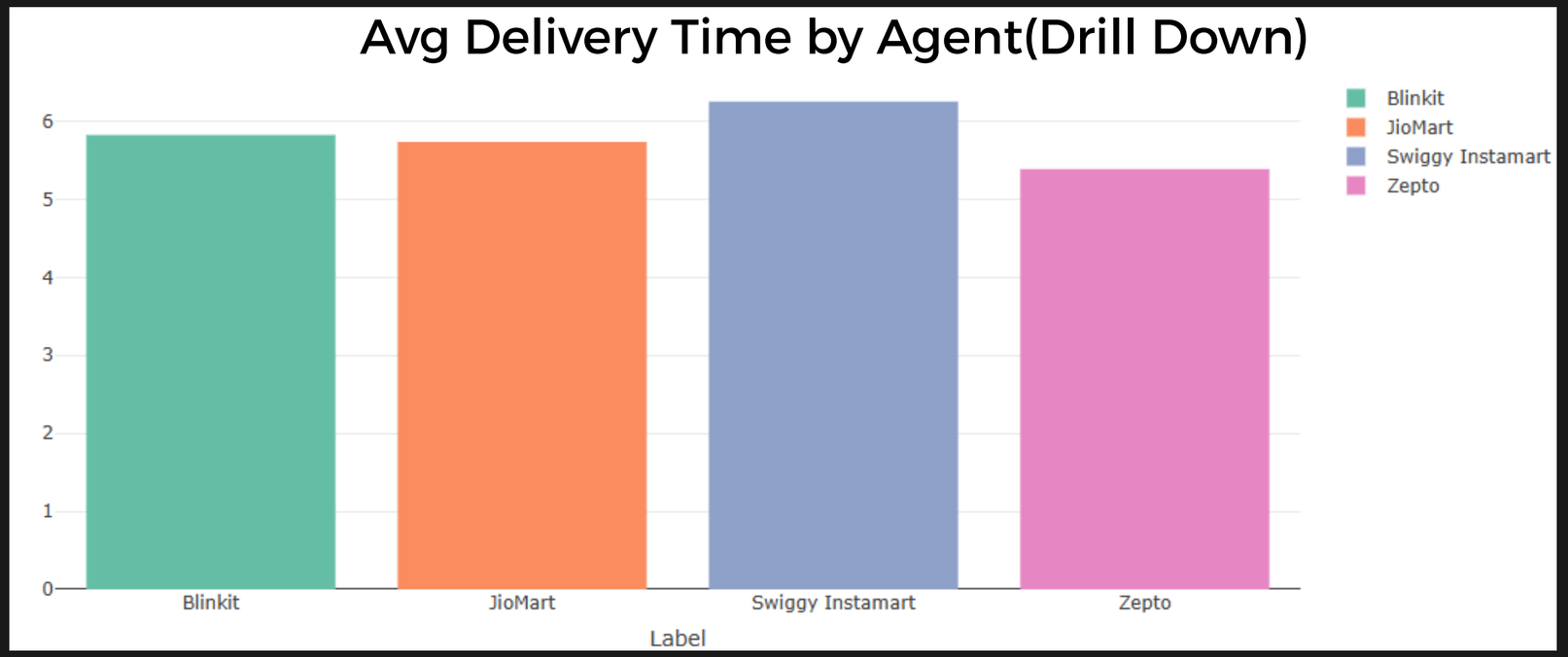
Dashboard: Shiny

Platform Overview: Delivery Trend



- Shows daily order volume over the analysis period
- Visible peaks and troughs indicate demand patterns (weekday vs weekend)
- Sudden dips may indicate weather disruptions or supply shortages





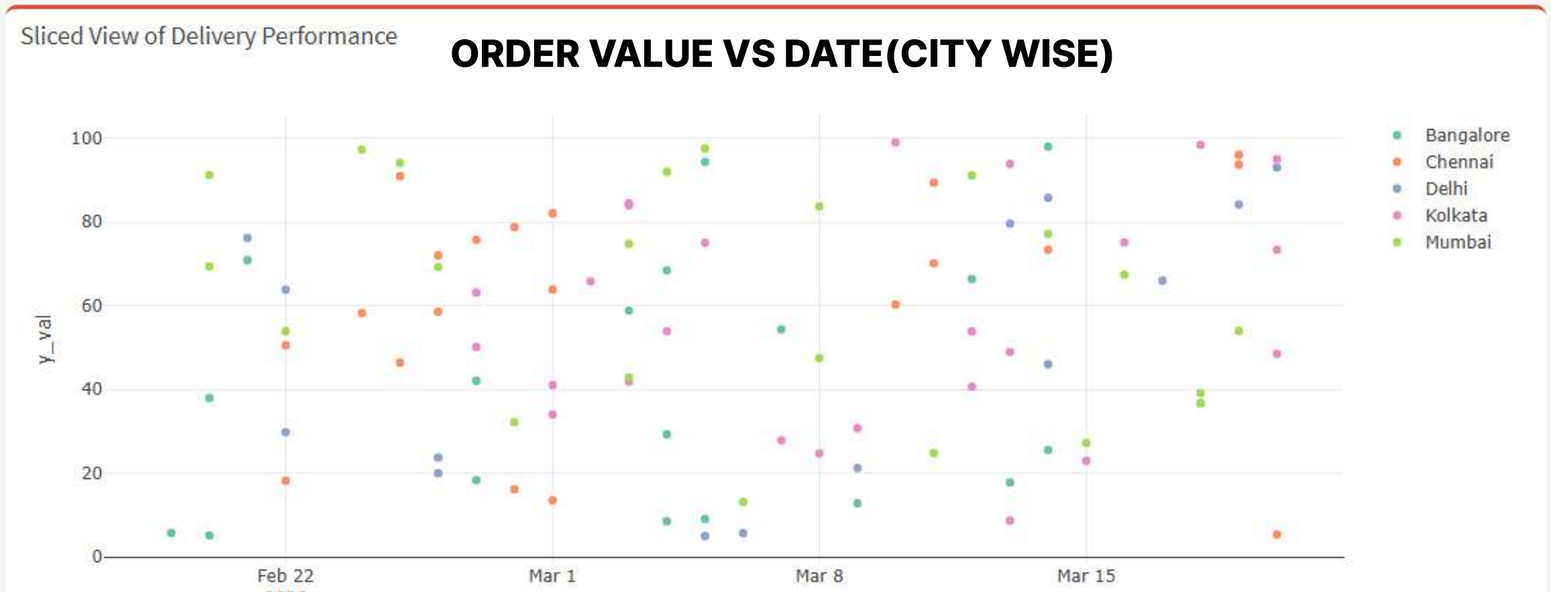
Roll-up aggregation: Average delivery time at city level

•FINDING: Bangalore = fastest delivery | Chennai = slowest delivery

•INFERENCE: City-specific logistics strategies needed, not one-size-fits-all

•Drill-down to agent level reveals individual performance bottlenecks

MULTI-DIMENSIONAL & SLICE/PICK ANALYSIS



- SCATTER: DISTANCE VS DELIVERY TIME, COLORED BY DELAY STATUS
- POSITIVE CORRELATION EXISTS, BUT DELAYS OCCUR EVEN AT SHORT DISTANCES
- SLICE & PICK ENABLES DYNAMIC FILTERING BY TRAFFIC LEVEL.

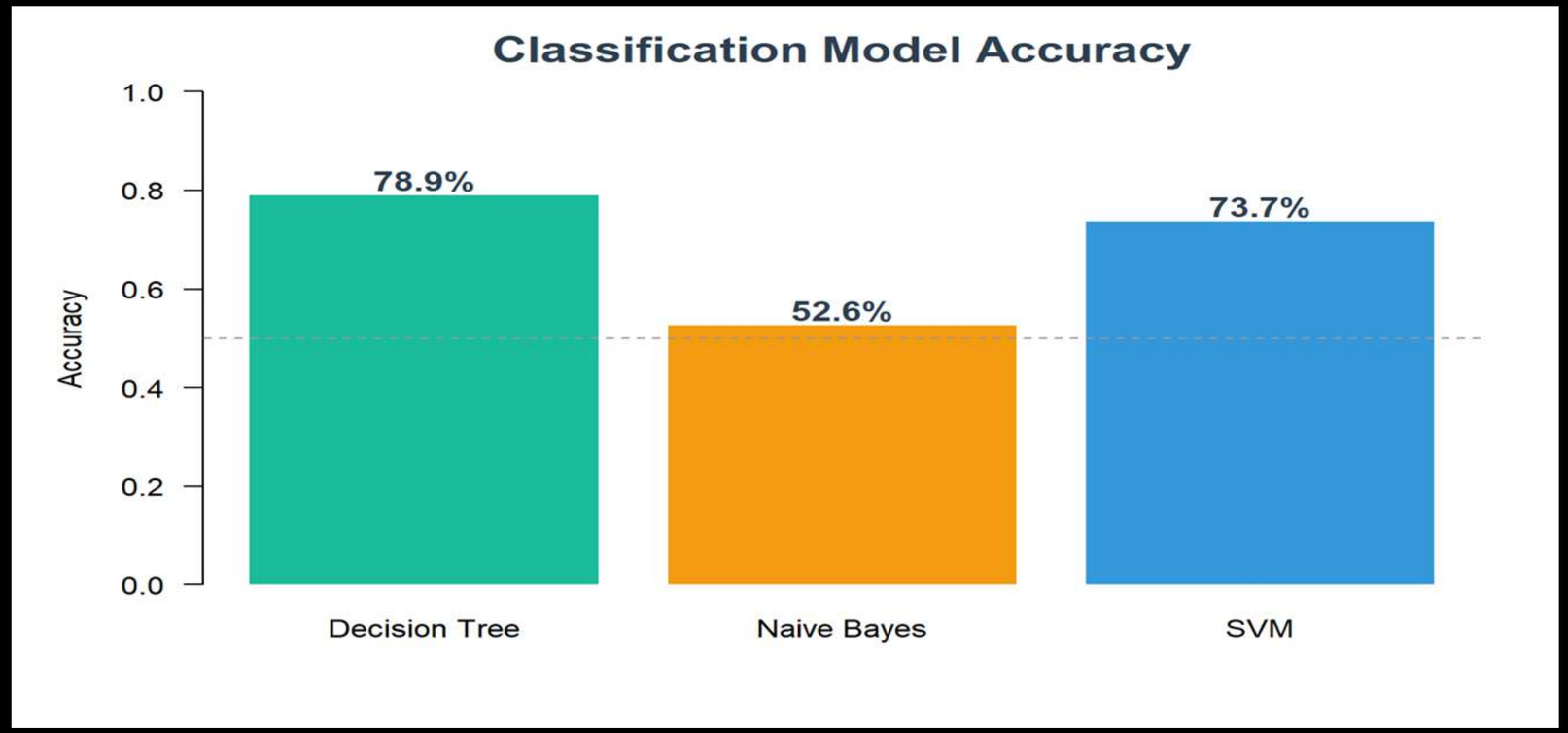
PCA

- PCA REDUCES MULTI-DIMENSIONAL FEATURES INTO 2 INTERPRETABLE COMPONENTS

- PCA REVEALS HOW TRAFFIC DENSITY AND DISTANCE CLUSTER IN THE FEATURE SPACE

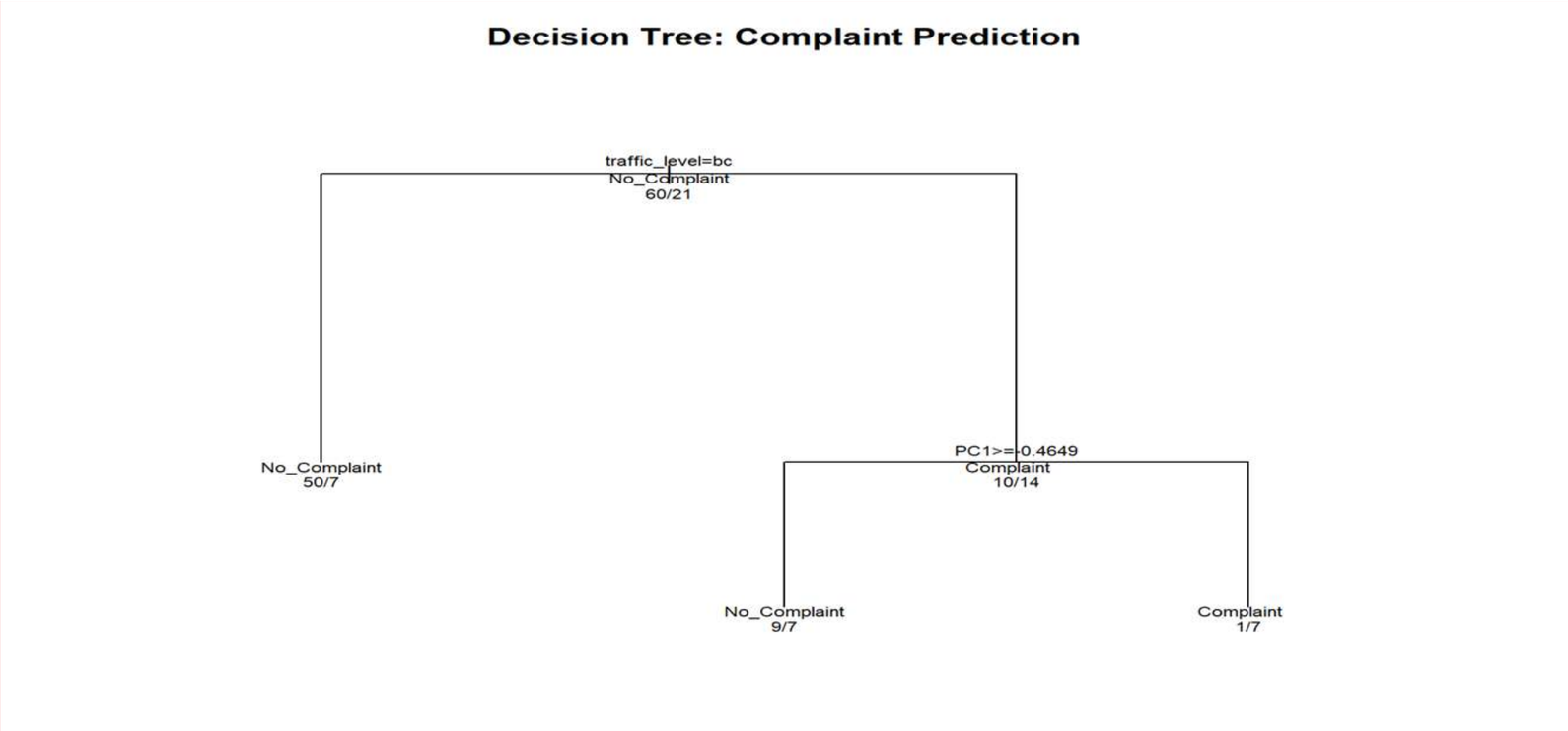
- INFERENCE: DIMENSIONALITY REDUCTION IS ESSENTIAL BEFORE CLUSTERING/CLASSIFICATION

CLASSIFICATION: MODEL PERFORMANCE



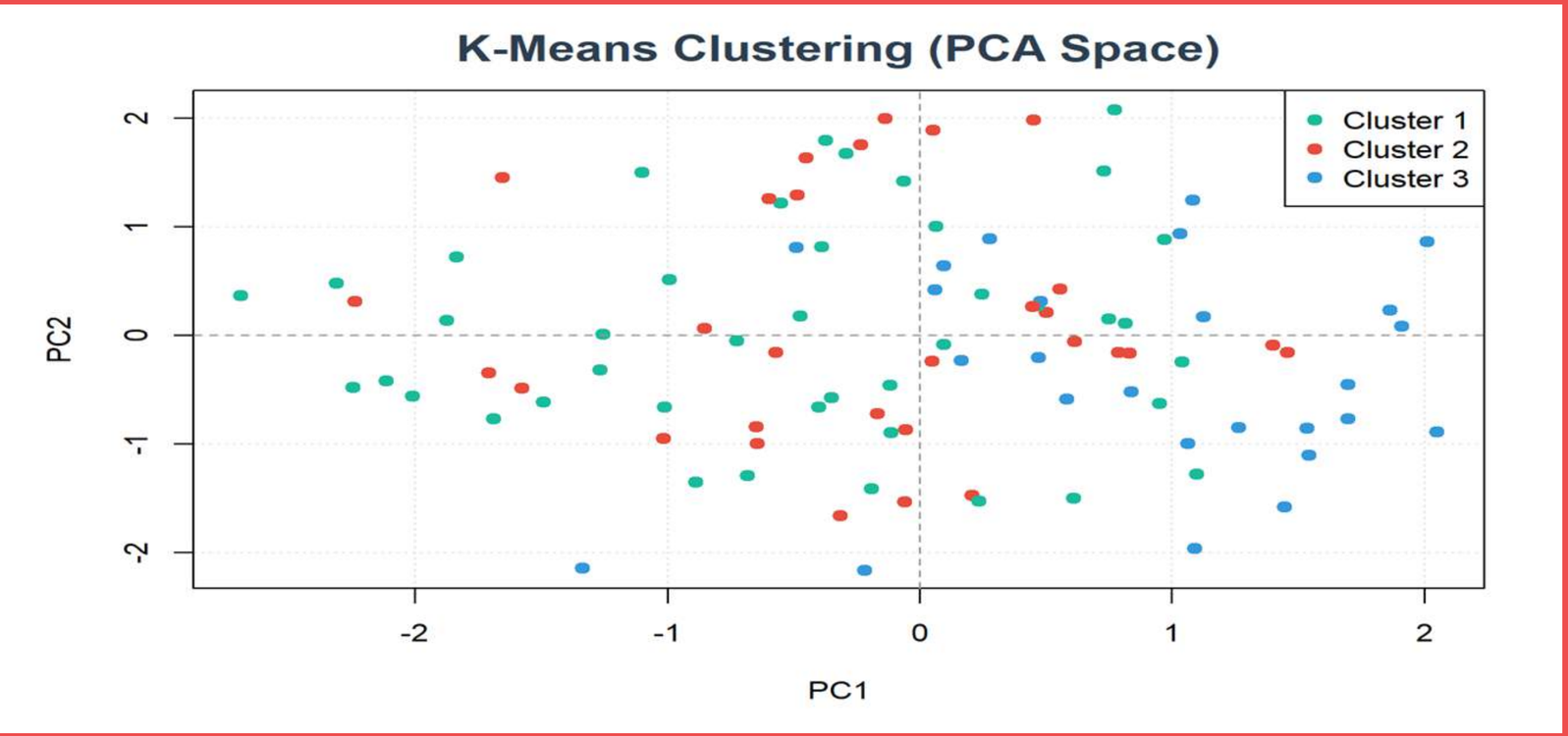
- THREE MODELS COMPARED: DECISION TREE, NAIVE BAYES, SVM
- BEST MODEL: DECISION TREE AT 78.9% ACCURACY FOR COMPLAINT PREDICTION
- INFERENCE: NON-LINEAR BOUNDARIES (DT/SVM) OUTPERFORM NAIVE BAYES ASSUMPTIONS
- DEPLOYING THE BEST MODEL ENABLES PROACTIVE CUSTOMER COMPLAINT PREVENTION

DECISION TREE: COMPLAINT PREDICTION



- TRAFFIC_LEVEL IS THE ROOT PREDICTOR - MOST IMPORTANT FEATURE
- HIGH TRAFFIC -> DELAYS -> HIGHER COMPLAINT PROBABILITY
- INFERENCE: PROVIDES INTERPRETABLE BUSINESS RULES FOR OPERATIONAL ALERTS
- RULE: IF TRAFFIC=HIGH AND DISTANCE>THRESHOLD THEN PROACTIVELY NOTIFY CUSTOMER

CLUSTER & OUTLIER ANALYSIS

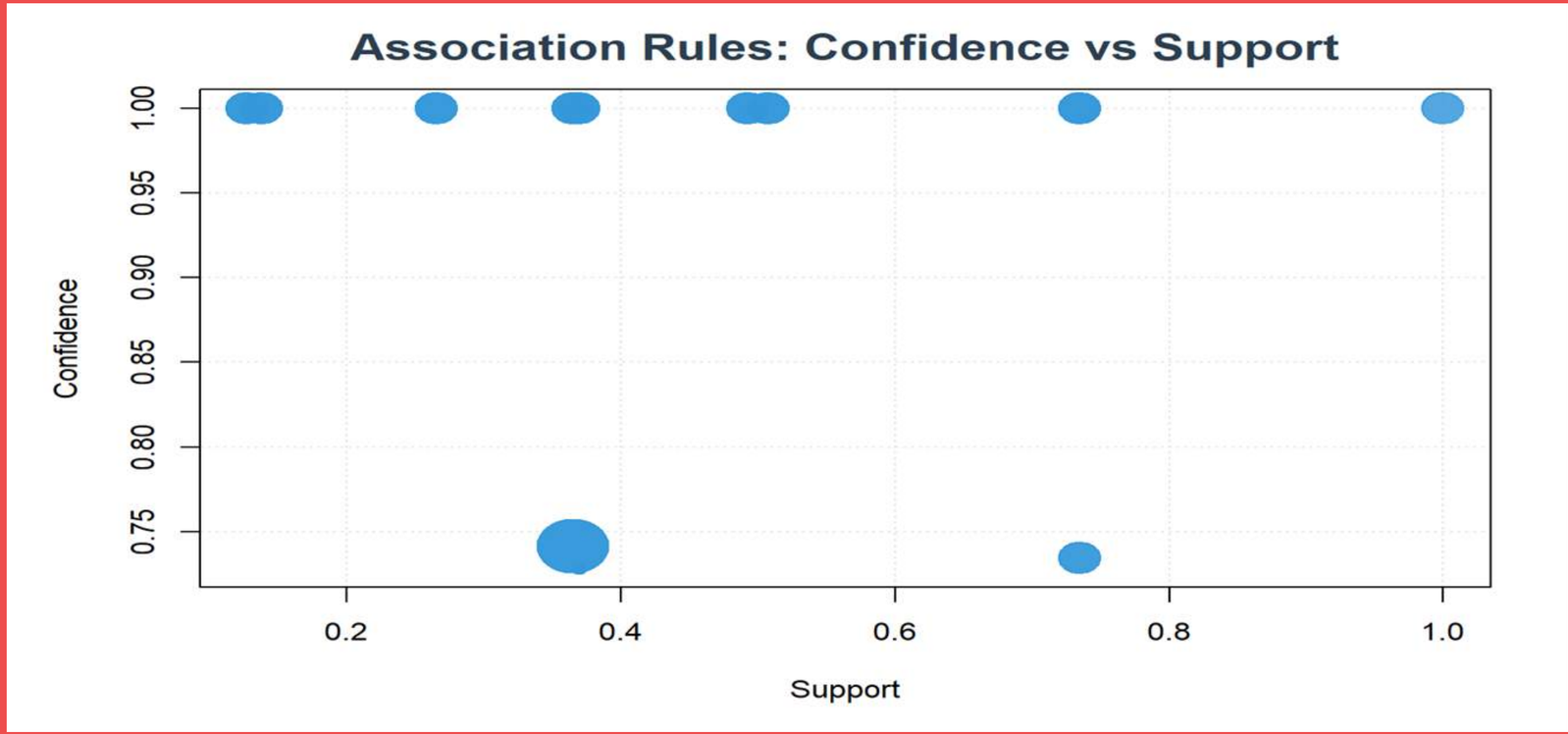


- K-MEANS REVEALS NATURAL DELIVERY BEHAVIOR SEGMENTS (K=3)

- CLUSTER 1 IDENTIFIED AS HIGHEST-RISK FOR DELIVERY DELAYS
- OUTLIER DETECTION: 5 ANOMALOUS DELIVERIES FLAGGED

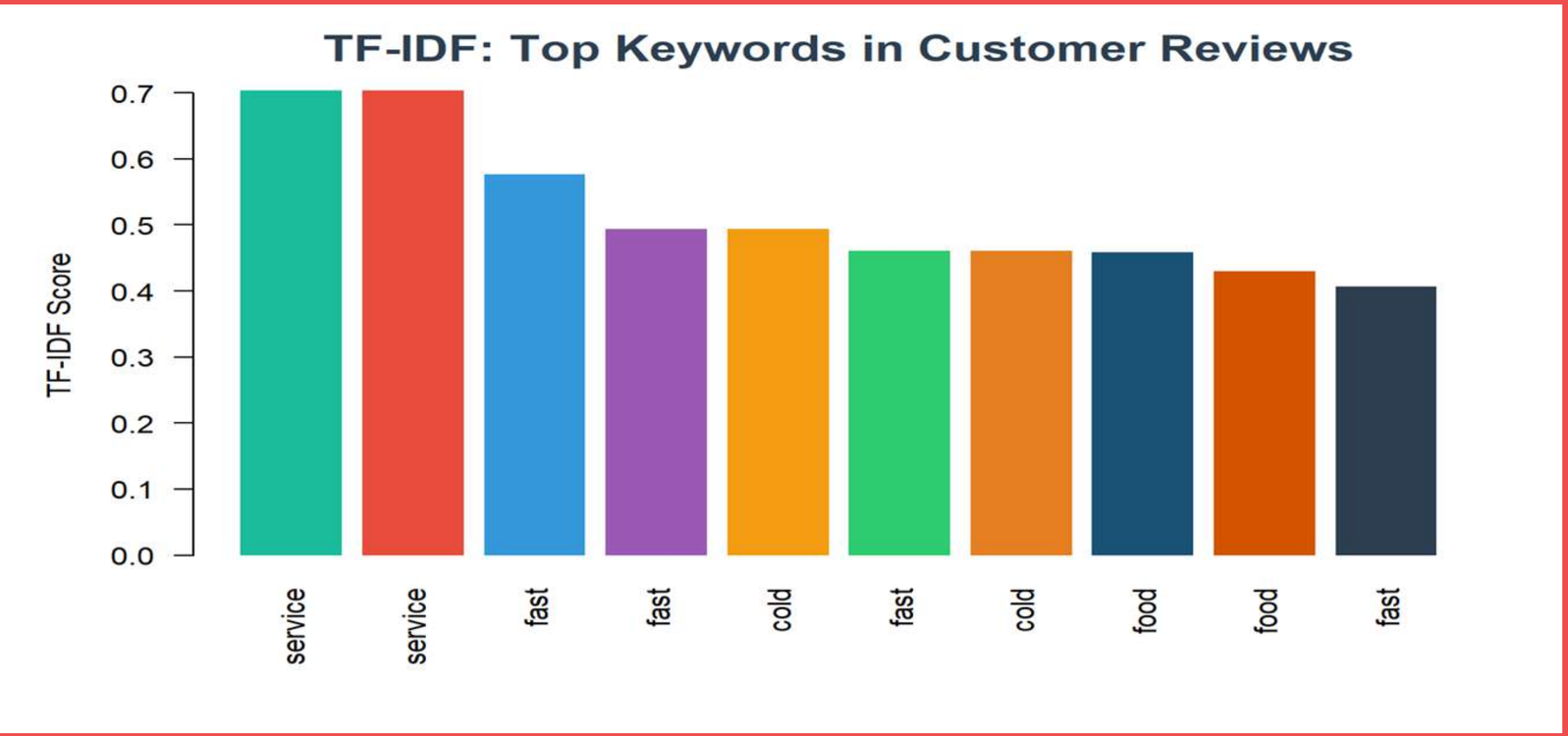
- INFERENCE: SEGMENTED STRATEGIES FOR EACH CLUSTER + AUTOMATED OUTLIER ALERTS

ASSOCIATION RULE MINING (APRIORI)



- 45 ASSOCIATION RULES DISCOVERED FROM DELIVERY AND REVIEW DATA
- KEY: {NO DISCOUNT} => {LOW RATING} - DISCOUNTS CRITICAL FOR SATISFACTION
- KEY: {LOW AVAILABILITY} => {LOW RATING} - AGENT SCHEDULING MATTERS
- INFERENCE: IMPLEMENT TARGETED DISCOUNTS & IMPROVE PEAK-HOUR AGENT SCHEDULING

TEXT MINING & WEB MINING INTELLIGENCE



- TF-IDF REVEALS TOP CUSTOMER REVIEW KEYWORDS: SPEED, FOOD QUALITY, SERVICE
- WEB MINING: 'FAST DELIVERY' AND 'CHEAP FOOD' DOMINATE SEARCH TRENDS
- INFERENCE: SPEED AND PRICE ARE PRIMARY CUSTOMER CONCERNS
- 'LATE NIGHT SNACK' SEARCHES HIGHLIGHT DEMAND FOR EXTENDED SERVICE HOURS

CONCLUSION & RECOMMENDATIONS

- DYNAMIC SERVICE AGREEMENTS: AUTOMATICALLY ADJUST "PROMISED DELIVERY TIME" BASED ON REAL-TIME TRAFFIC & WEATHER PREDICTIONS (SVM/NAIVE BAYES).
- PRODUCT QUALITY CONTROL: INVEST IN THERMAL PACKAGING TO ELIMINATE "COLD FOOD" COMPLAINTS IDENTIFIED VIA TEXT MINING (TF-IDF).
- PROACTIVE STAFFING: INCREASE DELIVERY FLEET DURING FESTIVAL CLUSTERS WHERE DELAY RISK IS PREDICTED TO BE 40% HIGHER.
- TARGETED RETENTION: LAUNCH "PRIORITY PASS" OFFERS FOR HIGH-VALUE/HIGH-DELAY CITIES IDENTIFIED THROUGH CLUSTERING (K-MEANS).
- SALES MAXIMIZATION: USE ASSOCIATION RULES (APRIORI) TO CREATE BUNDLED MEAL DEALS, INCREASING THE AVERAGE ORDER VALUE (AOV).

Thank You!